The paper “Regional and Local Economic Effects from Proximity of High-Speed Rail Stations in Japan: Difference-in-Differences and Propensity Score Matching Analysis” by Jetpan Wetwitoo and Hironori Kato explores the effect of proximity to high speed rail on the development of local and regional economies in Japan. Governments around the world invest billions of dollars into building high speed rail because it’s a relatively cheap and environmentally friendly method of long distance transport. However, building and maintaining it can be extremely costly, so it must be justified by a very strong projected economic benefit. Japan has been one of the most successful nations in the development of their high speed rail network, but it is still an open question as to how much this benefits their economy, as studies of their economic impact have produced mixed results. This study focuses on the economic impact of their newer extensions to their existing HSR network.

The study is based on data from the Statistics Bureau of Japan. The analysis was divided into two parts: analysis at the prefecture level, which is roughly similar to the state level in the U.S., and at the municipality level, which is similar to the county level in the U.S. The treatment group included all prefectures or municipalities in or near which one of the HSR lines they were studying was constructed. There were two control groups, one in which there were no HSR services during the entire study period (Control Group 1) and another in which the HSR network already covered the region (Control Group 2). Only Control Group 1 was used for the municipality analysis. The outcome variable for the prefecture level analysis was gross regional product (GRP), a measure of the total value of the goods and services produced in a region. The outcome variable for the municipality level analysis was taxpayer revenue, presumably because municipalities are much smaller than prefectures so GRP would be less reliable. The covariates used for the prefecture level analysis were labor input and capital input, while those for the municipality level analysis included the number of taxpayers, size of the municipality, and various characteristics of the taxpayer base. A 1:1 matching model was fit for the prefecture analysis. The results for the prefecture level analysis showed that there was a significant positive ATT for the treatment group relative to Control Group 1 at the $p<0.05$ level, but not at the $p<0.01$ level, and the ATT for Control Group 2 is not significant at any level. In addition, for both of these estimates, a difference in differences (DID) regression model was fit to estimate the change in their respective outcome variables over time. The DID was not significant at any level for both the prefecture based model or for any version of the municipality based model.

Ultimately, these results show little support that the new developments of HSR directly impacted local and regional economies. However, one limitation the authors note is that it may take longer than the study period for potential effects to be observable.